

BIG DATA ANALYTICS LAB

IV B. TECH. - I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5CS18	PCC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	30	70	100

COURSE OBJECTIVES

1. To introduce the terminology, technology and its applications
2. To introduce the concept of Analytics for Business
3. To introduce the tools, technologies & programming languages this is used in day to day analytics cycle.

COURSE OUTCOMES

1. Connect to hadoop cluster, **experiment** with various Linux and HDFS commands to store data.
2. **Apply** the knowledge of MapReduce programming to process the stored data in HDFS.
3. **Make use of** database operations to store results in tables and generate reports.
4. **Connect** to web data sources for data gathering, **Integrate** data sources with hadoop components to process streaming data.
5. **Generate** reports using data visualization tools.

LIST OF EXPERIMENTS:

WEEK 1

- i) Perform setting up and installing Vmware for Hadoop and Linux.
- ii) Basic Linux Commands

WEEK 2

Run basic HDFS shell commands

WEEK 3

Implement the following file management tasks in Hadoop:

- Adding files and directories
- Retrieving files
- Deleting files and directories.

Hint: A typical Hadoop workflow creates data files (such as log files) elsewhere and copies them into HDFS using one of the above command line utilities.

WEEK 4

Write the steps to export JAR using eclipse.

Run a basic Word Count Map Reduce program to understand Map Reduce Paradigm

WEEK 5

Write a Map Reduce program that mines weather data.

(Weather sensors collecting data every hour at many locations across the globe gather a large volume of log data, which is a good candidate for analysis with MapReduce, since it is semi structured and record-oriented).

WEEK 6

Run Pig and perform basic PIG commands.

WEEK 7

Write Pig Latin scripts to sort, group, join, project, and filter your data.

WEEK 8

Run HIVE and perform basic HIVE commands to create a table and enter data into tables.

WEEK 9

Use Hive to create, alter, and drop databases, tables, views, functions, and indexes

WEEK 10

Use CDH and HUE to analyze data and generate reports for sample datasets

WEEK 11

Importing and exporting Data in HDFS using Sqoop from MySql database

WEEK 12

Use data visualization tool to generate reports on sample datasets

TEXT BOOKS

1. Hadoop: The Definitive Guide, 4th Edition - O'Reilly Media
2. Understanding Big data , Chris Eaton, Dirk derooset al. , McGraw Hill, 2012.

REFERENCE BOOKS

1. Big Data Analytics, Seema Acharya, Subhasini Chellappan, Wiley 2015.
2. Intelligent Data Analysis, Michael Berthold, David J. Hand, Springer, 2007.
3. Harness the Power of Big Data The IBM Big Data Platform, Paul Zikopoulos ,Dirk DeRoos , Krishnan Parasuraman , Thomas Deutsch , James Giles , David Corigan , Tata McGraw Hill Publications, 2012.

WEB LINKS

1. https://onlinecourses.nptel.ac.in/noc20_cs92/preview
2. <https://nptel.ac.in/courses/110/106/110106064/>